MAT2040

Tutorial 4

TA Group

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June 17&18, 2024

Quick review

1. Computing matrix inverse
2. Permutation matrix and triangle matrix
3. LU decomposition and LDU decomposition

Question 1

A and B are $n \times n$ matrices. Prove that AB is invertible if and only if A and B are both invertible.

Solution

Prove if:

Since A, B are both invertible, A^{-1}, B^{-1} exist

$$(AB)(B^{-1}A^{-1}) = I$$
$$(AB)^{-1} = B^{-1}A^{-1}$$

AB is invertible.

Prove only if:

AB invertible $\rightarrow (AB)^{-1}$ exists $\rightarrow A[B(AB)^{-1}] = I$, which means A invertible and $A^{-1} = B(AB)^{-1}$

Similarly, $(AB)^{-1}AB = I \rightarrow$ B invertible, $B^{-1} = (AB)^{-1}A$

Question 2

Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}.$$

Calculate elementary matrices E_3, E_2, and E_1 that transform A into upper triangular form $E_3E_2E_1A = U$.

Solution

$$
\begin{aligned}
&E_3E_2E_1A \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = U
\end{aligned}
$$

Question 3

Find the $PA = LU$ factorization for $A = \begin{bmatrix} 0 & -1 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & 5 \end{bmatrix}$.

Solution

Let $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $PA = \begin{bmatrix} 2 & 1 & 2 \\ 0 & -1 & 3 \\ 4 & 3 & 5 \end{bmatrix}$.

Let $L_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$, then $L_1 PA = \begin{bmatrix} 2 & 1 & 2 \\ 0 & -1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$.

Let $L_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, then $L_2 L_1 PA = \begin{bmatrix} 2 & 1 & 2 \\ 0 & -1 & 3 \\ 0 & 0 & 4 \end{bmatrix} = U$.

Thus $L_2 L_1 PA = U$, $PA = L_1^{-1} L_2^{-1} U = LU$, where

$$L = L_1^{-1} L_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix}.$$

Question 4

Find the $PA = LDU$ factorizations (and check them) for

1) $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix}$

2) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix}$

Solution (1)

Start with $L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Firstly, we need to exchange the row 1 and row 2 of matrix A by using the permutation matrix, in this case the matrix P:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Then we apply the elementary row operations on the permuted matrix, and track the multipliers to obtain L:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \xrightarrow{R_3 \to -2 * R_1 + R_3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 2 \end{bmatrix}$$

$$\xrightarrow{R_3 \to -3 * R_2 + R_3} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

At this point, we already know the matrix L and P as:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We then need to determine the D and U. We divide each row of the final transformed matrix of A by diagonal entry in that row to produce U, and the divisor then going into D, that:

$$U = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Finally, to confirm the factors are correct, we compute the products LDU and PA:

$$PA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

$$LDU = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix} = PA$$

Solution (2)

Similarly, we can obtain the L, D, U and P matrix as below:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

And you can also compute the PA and LDU to confirm the factoriazation is correct.